

The shape became bounded, but it did not become dissipative

This release divides the localized filtered-vorticity covariance by its own trace to obtain a dimensionless shape variable. The exact evolution shows that a frozen resolved-strain source produces a nonnegative variance and drives alignment. In an exact periodic Navier–Stokes shear whose nonlinear term vanishes identically, the normalized viscous evolution takes both positive and negative signs. Trace normalization therefore cannot by itself supply the dissipative mechanism missing between Leray energy and critical control.

Status · R0.70K exact algebra, independent audit, and formal figure complete

The normalization-monotonicity route is closed; the compensator
is the next gate

Version v0.70K · 2026-08-24

Symbolic checks: all 6 groups pass

Independent mathematical audit: PASS

Literature boundary: 10 primary sources

Figure data: closed-form formulas, not DNS

00 / EXACT CONCLUSION

Bounded does not mean dissipative

Such that

$$Q = \int \chi \Omega \otimes \Omega \, dx, \quad E = \operatorname{tr} Q > 0,$$

and define the trace-one shape $R = Q/E$ and the dimensionless anisotropy

$$B = R - \frac{1}{3}I = \frac{\operatorname{dev} Q}{E}.$$

Positive semidefiniteness gives the sharp bounds

$$-\frac{1}{3} \leq \lambda_i(B) \leq \frac{2}{3}, \quad 0 \leq |B|_F^2 \leq \frac{2}{3}.$$

But bounded does not mean small, much less dissipative. Rank-one vorticity attains $2/3$ exactly, while the raw stretching work still contains the localized vorticity mass E , which normalization does not remove.

The exact negative conclusion of this release. A frozen source makes the signed correlation q nondecreasing, while pure diffusion after normalization can either increase or decrease $|B|_F^2$. Trace normalization alone does not automatically produce a universal monotonicity law.

01 / NORMALIZED COVARIANCE IDENTITY

The genuinely normalized object is $B = A/E$

The raw deviatoric moment $A = \text{dev } Q$ proposed at the end of Ro.70J is still dimensional. This release explicitly replaces it with $B = A/E$; I do not rewrite the previously hash-sealed report.

If $F = \dot{Q}$, direct differentiation gives

$$\dot{B} = \frac{\text{dev } F - B \text{tr } F}{E}.$$

When $F = \lambda Q$, the right-hand side vanishes exactly, so a pure amplitude change is removed exactly. The complete filtered-covariance flux decomposes as

$$\dot{Q} = F_\chi + F_S + F_\nu + F_C,$$

These terms represent transport through a moving or fixed cutoff, resolved-strain stretching, viscous diffusion, and subfilter exchange, respectively. Each term must carry its own denominator-trace correction; retaining only $\text{dev } F/E$ omits a term. Normalization is undefined at $E = 0$, and the sharp bound does not resolve the denominator problem when E is small.

02 / FROZEN-SOURCE VARIANCE LAW

A frozen source produces a nonnegative variance, not damping

For a spatially constant, symmetric, trace-free source Σ , let $q = \Sigma : B = \Sigma : R$. The source-induced shape equation is

$$\dot{R}|_\Sigma = \Sigma R + R \Sigma - 2qR.$$

The correlation obeys the exact variance law

$$\dot{q}|_{\Sigma} = 2 \operatorname{tr} [R(\Sigma - qI)^2] \geq 0.$$

Equality holds only when the support of R lies entirely in one eigenspace of Σ . A frozen resolved strain does not dissipate its correlation with the vorticity shape; it drives the covariance toward an eigendirection of the source.

For $\Sigma_0 = \operatorname{diag}(-1/2, -1/2, 1)$, starting from isotropy gives explicitly

$$q(t) = \frac{e^{3t} - 1}{e^{3t} + 2}, \quad |B(t)|_F^2 = \frac{2}{3} q(t)^2.$$

The correlation increases monotonically from zero toward the maximal extensional value of unity, while the anisotropy approaches the sharp rank-one bound.

03 / EXACT PERIODIC SHEAR

The normalized viscous term takes both signs

On the three-dimensional periodic domain, consider

$$u(t, z) = \left(\frac{D}{2} e^{-4\nu t} \sin 2z, -C e^{-\nu t} \sin z, 0 \right).$$

The velocity depends only on z and has zero vertical component, so $(u \cdot \nabla)u = 0$. The vorticity also has zero vertical component, so the stretching term vanishes as well. The two modes decay only at the rates of their respective heat equations.

Take the identity filter and the full-torus average with $\chi \equiv 1$. Writing the normalized vorticity covariance as $R = \operatorname{diag}(p, 1 - p, 0)$ gives

$$\frac{d}{dt} |B|_F^2 = 12\nu p(1 - p)(2p - 1).$$

At $p = 4/5$, the derivative is $+144\nu/125$; at $p = 1/5$, it is $-144\nu/125$. This sign pair contains no nonlinearity, pressure, cutoff, or subfilter exchange. It directly proves that a negative-semidefinite raw diffusion matrix does not produce universal anisotropy monotonicity after trace normalization.

04 / BURGERS-VORTEX BOUNDARY

The complete equations can preserve maximal anisotropy

The classical Burgers vortex provides a self-consistent Navier–Stokes boundary on the other side. Its background strain is

$$\Sigma_\gamma = \gamma \operatorname{diag}(-1/2, -1/2, 1),$$

The axial Gaussian vorticity gives

$$R = e_3 \otimes e_3, \quad B = \operatorname{diag}(-1/3, -1/3, 2/3), \quad q = \gamma > 0.$$

Here B is already at the maximal-anisotropy vertex, yet it remains stationary. Transport and viscosity in the complete steady equations exactly compensate the positive raw stretching.

Boundary. The classical Burgers vortex has a linearly growing background velocity and an infinitely long axial vortex tube, so it is not a finite-energy Leray solution on $\mathbb{R}^3$. It is a structural witness for the complete equations, not a counterexample to regularity.

05 / EXACT COMPENSATOR LEDGER

The missing mechanism is compressed into one explicit ledger

Write the actual resolved strain as the sum of a constant reference source and a residual, and let F_{err} collect the nonconstant strain, transport, cutoff, diffusion, and subfilter exchange terms. This gives

$$\dot{q} = \dot{\Sigma} : B + 2 \operatorname{tr}[R(\Sigma - qI)^2] + \Sigma : \mathcal{T}_B(F_{\text{err}}).$$

The middle term is known to be nonnegative. Any possible control must come from the source evolution itself or from joint compensation by the residual fluxes. For an actual low-frequency source, $\dot{\Sigma}$ also brings in the deviatoric pressure Hessian, strain self-amplification, vorticity, viscosity, and the subfilter Hessian. These terms cannot be replaced by a local eddy viscosity without proof.

Therefore, my next step is not to seek another sign for B alone. I will evolve (E, B, Σ) jointly under the same filtering and cutoff conventions, and then check whether compensation appears after summing adjacent scales.

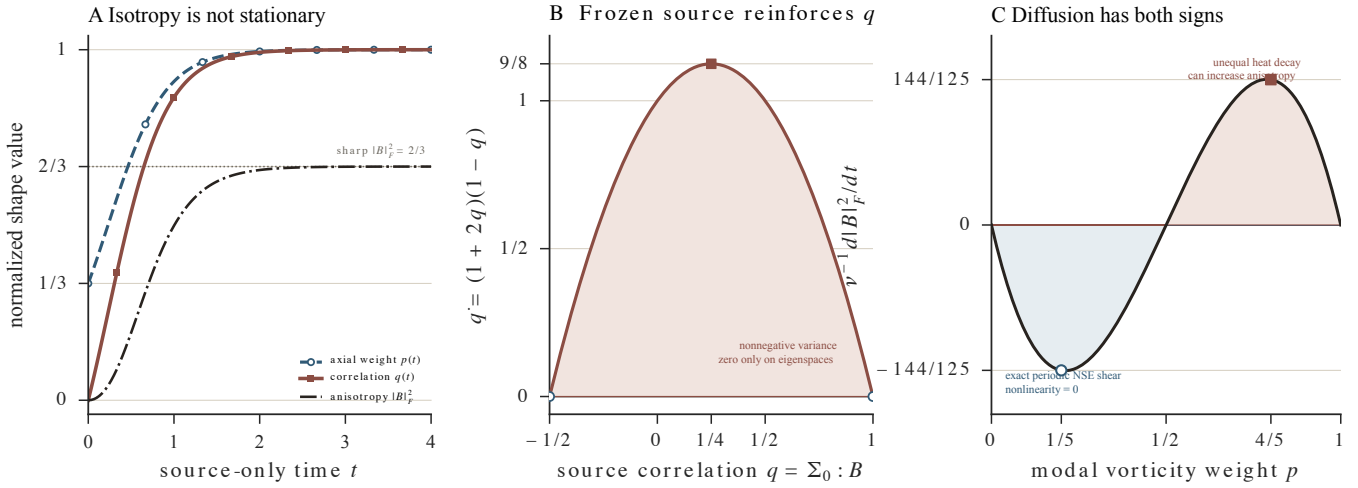
Closed-form trajectories, nonnegative variance, and a viscous sign pair

Normalized vorticity anisotropy: alignment and the missing sign

frozen-source covariance flow · exact variance law · periodic Navier–Stokes diffusion gate



EXACT ANALYTIC IDENTITIES / NOT DNS / NOT A REGULARITY OR BLOW-UP RESULT



closed-form covariance identities — full source evolution and Leray endpoint control remain open

Figure Ro.70K. A: Under a frozen axisymmetric source, an isotropic covariance approaches a rank-one state. B: The source-correlation production $(1 + 2q)(1 - q)$ is nonnegative on the realizable interval. C: The exact periodic two-mode shear gives positive and negative regions of $\nu^{-1} d|B|_F^2/dt = 12p(1 - p)(2p - 1)$. Every curve comes from a closed-form formula, not DNS.

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07 / RESEARCH VALUE AND LITERATURE BOUNDARY

The value is to close one route and leave a testable new gate

Ro.70K is not a regularity theorem. It uses the frozen-source identity and the exact periodic shear to rule out the tested universal monotonicity, while the Burgers vortex marks the compensation boundary of the complete equations. There is no longer sufficient reason to continue studying an algebraic sign of B alone.

A bounded search of ten primary sources found exact filtering identities, conditional scale locality, averaged flux theorems, and statistical closures, but no deterministic sign estimate for the fixed-cutoff, trace-normalized vorticity covariance used on this page. This states the boundary of the search protocol; it is not a proof that no such literature exists.

Next, I will examine only source-evolution compensation: derive the actual $\dot{\Sigma}_k$, retain the deviatoric pressure Hessian and cross-scale flux, and then test whether an adjacent-scale combination can compensate the nonnegative variance. If the sign still fails, I will first construct an exact witness and only then decide whether a large simulation is needed.

The conclusion applies only at the stated smooth, fixed-filter gate

PROVED Normalization and variance identities The six groups of exact checks agree with the independent derivation.	PROVED Periodic viscous sign pair An exact Navier–Stokes heat shear isolates the diffusion term.
OPEN Denominator and weak-solution limit $E = 0$, low-mass regions, and the Leray limit are not treated.	OPEN Source-evolution compensation The joint pressure and adjacent-scale ledger is not yet closed.

The identities apply to smooth solutions, a fixed filter that commutes with spatial derivatives, a smooth cutoff, and times at which $E > 0$. A time-varying filter scale, a sharp cutoff, and the weak-solution limit require additional arguments.

The periodic two-mode shear is not a finite-energy cascade on $\mathbb{R}^3$, and the Burgers vortex is not a finite-energy Leray field either. This release proves neither a finite-time singularity nor global regularity, and it does not solve the Millennium problem.

The exact producer, independent audit, literature boundary, and figure package can be checked separately

[View the exact certificate](#) · [View the complete mathematical report](#) · [View the primary-literature audit](#) · [View the independent audit](#)

[View the ten-file formal figure package](#) · [下载同步 PDF \(Chinese PDF\)](#) · [View the Clay Mathematics Institute's official problem statement](#)

The certificate producer exactly checks normalization, the frozen source, the axisymmetric model, realizability, the scalar Burgers balance, and the periodic shear. General positivity for an arbitrary positive-semidefinite R , the complete Burgers velocity-pressure solution, the weak-solution limit, and completeness of the literature search remain separate analytic claim boundaries.